

END-TO-END DEMONSTRATION

AI for data, writing & presentation.

From a spreadsheet to a manuscript to a conference deck, with the right tool at each step and your judgement at every checkpoint.

ANALYSE

WRITE

PRESENT

INTEGRITY

Dr. Abhishek J. Benur · Dr. Saikat Banerjee

Pulmonologist · Co-Founder, Parabox AI · Author, The 1,000 Medical Prompts

@abhishekjbenur

PULMOCON 2026 · 8:40 PM

THE PIPELINE

One dataset, **five stops**

01**Clean and de-identify**

Remove identifiers; build a data dictionary.

›

02**Analyse**

Plan the tests, run them in chat, confirm in a stats package.

›

03**Tables and figures**

Publication-ready Table 1, results tables, charts from code.

›

04**Write**

IMRaD draft from your results; you own the argument.

›

05**Present**

Abstract, poster and slides from the same verified content.

At every arrow: **a human checkpoint**. AI speeds each stop; you sign off before the next.

01

DATA

Statistics **without the syntax**

AI can write and run the analysis code. You still choose the question and check the answer.

THE TOOLS

Where to **run the analysis**



RUNS REAL PYTHON

ChatGPT data analysis

Upload a CSV or Excel file; it writes and runs code you can inspect, and draws charts.



ANALYSIS AND FILES

Claude

Runs code on uploaded data and builds real Excel, Word and PowerPoint files from the results.



PURPOSE-BUILT

Julius AI

Chat-to-chart for larger datasets; Python and R behind a simple interface.



FREE · VALIDATE HERE

JASP · Jamovi · R

Point-and-click, publication-grade output. Confirm every key result here.

Upload **de-identified data only**, and check your ethics approval allows it to leave the institution.

DEMO PROMPT

From a CSV to a statistical analysis plan

PROMPT · UPLOAD THE DE-IDENTIFIED CSV

PERSONA: You are a biostatistician who supports MD thesis students and is strict about test assumptions.

MATERIAL: [De-identified CSV]. Research question: [Is eosinophil count ≥ 300 associated with 90-day readmission in COPD exacerbation?]. Design: [retrospective cohort, n=212].

TASK:

1. Describe the variables and missing data.
2. Propose the analysis plan: tests for each objective, and why.
3. Check the assumptions, then run the analysis; show the code.
4. Build Table 1 and the main results table (effect size, 95% CI, p).
5. List the limitations of this analysis.

RULES: Don't switch tests to get significance. Flag small cells and multiple comparisons. Write [CHECK IN JASP] next to each key result.

WHY IT WORKS

Plan before running

The analysis plan comes first.

Assumptions checked

The wrong test is the commonest AI error.

Code shown

You can reproduce and audit it.

Validation flag

Every key number is re-run in JASP.

BEFORE A NUMBER GOES IN YOUR PAPER

The analysis sanity check



Right test?

Matches the data type, distribution and design.



Effect size and CI

Not just a p-value: what the difference is and how sure.



Reproducible

Save the code or the JASP file with your data.



Re-run independently

The same result in JASP, Jamovi or R.



No p-hacking

Tests fixed in advance; no switching to get $p < 0.05$.



A statistician's eye

For your thesis or a paper, show it to one before submitting.

02

WRITING

AI as editor, not author

You bring the results and the argument; AI helps with structure, clarity and journal fit.

THE WRITING STACK

The right tool for each writing job

LONG-FORM DRAFTING

Claude

Structures IMRaD sections, tightens prose, matches a target journal's style.

FAST REWRITES

ChatGPT

Titles, abstract versions, cover letters, responses to reviewers.

CITE AS YOU WRITE

Jenni AI

Inline citations in Vancouver and other styles, from PubMed and Semantic Scholar.

ACADEMIC POLISH

Paperpal

Language editing and pre-submission checks built for manuscripts.

REFERENCES

Zotero

Free. Saves papers, formats citations, flags retracted papers.

CITATION CHECK

Scite

Does later work support or contradict what you're citing?

DEMO PROMPT

Draft the discussion from your results

PROMPT · PASTE YOUR RESULTS AND KEY PAPERS

PERSONA: You are a senior respiratory physician and journal reviewer who writes concise discussions.

MATERIAL: My results: [paste Table 2 and main findings]. Key papers (abstracts or PDFs): [paste 5-8]. Target journal: [Lung India].

TASK: Draft a 700-word discussion in this order:

1. Main finding in one sentence.
2. Comparison with each key paper: agree or disagree, and why.
3. Possible mechanisms.
4. Strengths, then limitations (honestly).
5. Implications for Indian practice and research.

RULES: Cite only the papers I pasted, as [Author, year]. Don't overstate causation from observational data. Mark any claim without a source [CITATION NEEDED].

WHY IT WORKS

Your evidence only

No invented references.

Fixed structure

The order reviewers expect.

Honest limits

Reviewers look here first.

[CITATION NEEDED]

Gaps are visible, not filled.

PUBLICATION INTEGRITY

The rules **journals now enforce**



ICMJE AND COPE

AI is never an author

- AI can't take responsibility, so it can't be listed as an author
- You are accountable for every sentence
- Disclose AI use as the journal requires (methods or acknowledgements)



DON'T DO THIS

Common violations

- Uploading a manuscript you're peer reviewing to a chatbot
- AI-generated images presented as data
- References you haven't read
- Paraphrasing tools used to hide copying

Check the target journal's AI policy **before** you start writing, not after acceptance.

03

PRESENTATION

From manuscript to slides and posters

The same verified content becomes an abstract, a poster and a talk, in a fraction of the time.

DECK-BUILDING ROUTES

Five ways to build the deck



Claude

Builds real, editable PowerPoint files. This deck was made this way.



Claude Design

A visual canvas: refine slides and posters by chatting.



Gamma

Fastest prompt-to-deck; connects to ChatGPT and Claude.



NotebookLM

Slides grounded only in your uploaded sources.



Copilot in PowerPoint

Generates inside PowerPoint for Office users.

Need a .pptx file for the venue? **Build in Claude or Copilot**, and test any web-tool export the day before.

DEMO PROMPT

Paper to conference talk

PROMPT · CLAUDE · ATTACH THE MANUSCRIPT

PERSONA: You are a medical educator who designs clear, visual conference talks.

MATERIAL: [Attached manuscript]. Talk: [8 minutes, Pulmocon free-paper session]. Audience: [pulmonologists and residents]. Style: [dark background, one accent colour, minimal text].

TASK: Build a [10]-slide PowerPoint:

1. Title; 2. Why this question matters (one statistic); 3. Objective; 4. Methods as a flow diagram; 5-7. One result per slide as a chart or table; 8. Limitations; 9. Take-home message; 10. Thank you and contact. Add speaker notes to every slide, about 50 seconds each.

RULES: Use only numbers from the manuscript. One idea per slide, 30 words maximum.

WHY IT WORKS

Time drives length

8 minutes means ~10 slides.

Slide-by-slide plan

No generic 'make slides'.

Speaker notes

Your script, timed.

Numbers from the paper

No drift in the data.

POSTERS AND FIGURES

Make the **visuals** earn their place



Posters

Ask for a 3-column layout with a one-line headline finding and a QR code to the full abstract. Refine in Claude Design or PowerPoint.



Charts from data

Generate charts with code from your actual dataset, never from an image model. Check the axes and the n.



Schematics

Image models are fine for concept diagrams and mechanisms, labelled as illustrations, never as data or clinical images.

The line you never cross: **an AI-generated image is never data.**

SAVE THIS SLIDE

The cheat sheet

STEP	PRIMARY TOOL	YOUR CHECKPOINT
Analysis plan and code	ChatGPT data analysis · Claude	Right tests, assumptions met
Validate the results	JASP · Jamovi · R	Same numbers, independently
Draft the manuscript	Claude · ChatGPT	Your argument, your voice
Citations and references	Jenni AI · Zotero · Scite	Every reference opened and read
Polish and submit	Paperpal	Journal AI policy and disclosure
Slides and poster	Claude · Claude Design · Gamma	Numbers match the paper

TAKE-HOME

Faster work. Same integrity.

AI can compress weeks of analysis, writing and slide-making into days. The question, the checks and the name on the paper stay yours.

UP NEXT

Using AI Safely: Check Before You Trust · interactive closing · 8:55 PM